

Syeda Rija Ali

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SUMMARY

Fascinated by how intelligence emerges from scale, data, and structure and how that shapes the way models reason and decide. Have built production-grade multi-agent systems, MLOps pipelines, and computer vision applications that go beyond demos into deployed reality. Building toward founding something AI-driven, starting from a deep understanding of what's happening inside the systems I build.

EDUCATION

FAST NUCES Karachi

B.S. Artificial Intelligence

Expected Graduation, May 2027

- o **Concentrations:** Artificial Intelligence & Machine Learning Systems and Engineering

SKILLS

Programming: Python, SQL, C++, MongoDB,

AI/ML: PyTorch, HuggingFace, scikit-learn, MLflow, LangChain, RAG, Prompt Engineering, Embeddings & Semantic Search, Multi-Agent Systems

Backend/Infra: FastAPI, Redis, Celery, PostgreSQL, Docker, AWS EC2, GitHub Actions

Frontend: React, HTML, CSS

Tools: Git, Jupyter, YOLO, OpenCV, Pandas, NumPy, Matplotlib

PROJECTS

GitHub

MediFlow - AI-Powered Clinic Operations Platform · Multi-Agent · RAG · MLOps · DevOps

Jan 2026 – Apr 2026

- o Built a production-deployed multi-agent healthcare platform with RAG-based clinic knowledge retrieval, a guarded ReAct booking agent (DB-verified appointments, no hallucinated confirmations), and real-time voice automation via Deepgram STT/TTS + WebSocket streaming.
- o Implemented a full MLOps loop - XGBoost predictions for wait-time and patient load, MLflow model registry, drift detection, automated retraining via Celery Beat, and champion/challenger promotion.
- o Deployed on AWS EC2 with Docker Compose, Caddy, Vercel frontend, GitHub Actions CI/CD, and Neon PostgreSQL - fully automated on push to main.

Stack: FastAPI, React/TypeScript, PostgreSQL, Redis, Celery, ChromaDB, MLflow, Gemini, Groq, Deepgram, Docker, AWS EC2, GitHub Actions

Vision-to-Voice - Navigation System for Visually Impaired Users · Vision · World Models · Multi-Agent

Feb 2026 – Apr 2026

- o Built a map-free, real-time navigation system using DINOv3 embeddings for semantic scene understanding, YOLOE for object detection, and EasyOCR for landmark recognition - designed for zero-infrastructure deployment.
- o Constructed a cognitive spatial model using JEPA-based world modeling for predictive reasoning about unseen space - moving beyond reactive detection toward anticipatory navigation.
- o Applied explainability techniques (SmoothGrad, AttnLRP) to make AI guidance decisions interpretable, enabling trust calibration in safety-critical navigation guidance.

Stack: PyTorch, DINOv3, JEPA, YOLOE, EasyOCR, OpenCV, NetworkX

LAYR - ML-Powered Fashion E-commerce System · Recommendation · NLP · Time Series Forecasting

Aug 2025 – Dec 2025

- o Engineered an end-to-end ML pipeline over 80M+ records (5.2GB) of transactional, behavioral, and review data - covering ingestion, feature engineering, modeling, and evaluation.
- o Built a hybrid recommendation engine combining collaborative filtering, content-based similarity, and popularity signals; applied BERT-based NLP to extract sentiment and thematic signals from customer reviews.
- o Modeled product demand with Prophet and ARIMA for seasonal forecasting; performed RFM-based customer segmentation to support personalized marketing strategies.

Stack: Python, PyTorch, scikit-learn, Sentence-BERT, Prophet, PostgreSQL, FastAPI, MLflow

EXPERIENCE

Project Management Intern - AI Systems · [SysLab.ai](https://www.syslab.ai)

Nov 2025 – Feb 2026

- o Coordinated delivery workflows for AI-based assessment and proctoring platforms across planning, execution, and release phases.
- o Bridged technical and non-technical teams to sharpen requirement clarity and reduce timeline slippage across iterative feature releases.

Co-Founder · [wearrui](https://www.wearrui.com) (Clothing Brand)

Aug 2025 - Present

- o Co-founded a minimalist fashion brand; led brand strategy, visual identity, supplier coordination, and early-stage marketing positioning.